

Predicting Infections during hospitalization among postsurgical patients

A multimodal ML + NLP framework for early infection risk

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Hospital-acquired infection: the clinical problem

2 - 5 %

of operations develop a SSI^[1]

3.5 - 10 B\$

excess annual cost from SSIs
in the US alone ^[2]

0

national SSI surveillance
programs in Israel ^[3]

Today- detection of infection is usually confirmed only once it is already apparent.

^[1] World Health Organization. *Global Guidelines for the Prevention of Surgical Site Infection*, 2nd ed. Geneva: WHO, 2018.

^[2] Chakraborty A, et al. "Development and evaluation of machine learning models for the identification of surgical site infection in electronic health records." *Surgical Infections* 26, no. 7 (2025): 474.

^[3] Najjar Debbiny R, et al. "Healthcare-associated infection prevention and control practices in Israel: results of a national survey." *BMC Infectious Diseases* 22 (2022): 739.

The research gap

01

**Structured models
often lack external
validation^{[4][5]}**

02

**Text adds value - but
representation
matters^{[6][7]}**

03

**Hebrew multimodal SSI
prediction remains
unexplored^[8]**

The gap this project addresses: one evaluated, leakage-controlled system combining structured EHR data with Hebrew free text- not two models reported separately.

[4] Ndjongo LCM, et al. "Evaluating predictive performance and generalizability of traditional and artificial intelligence models in predicting surgical site infections post-spinal surgery: a systematic review." Spine Journal 26, no. 2 (2026): 280–291.

[5] Cui Y, Shi X, Wang Q, et al. "Artificial intelligence-based prediction model for surgical site infection in metastatic spinal disease: a multicenter development and validation study." International Journal of Surgery (2025).

[6] Bonde A, et al. "Assessing the utility of deep neural networks in detecting superficial surgical site infections from free text health record data." Frontiers in Digital Health 5 (2024): 1249835.

[7] Verberk JDM, et al. "The augmented value of using clinical notes in semi-automated surveillance of deep surgical site infections after colorectal surgery." Antimicrobial Resistance & Infection Control 12 (2023): 117.

[8] Saban M, et al. "Identifying diabetes related-complications in a real-world free-text electronic medical records in Hebrew using natural language processing techniques." Journal of Diabetes Science and Technology 19, no. 4 (2025): 99—1007.

Project goals

- 1 Develop a data-driven ML model to predict the risk of SSI during hospitalization.
- 2 Enable early identification of elevated-risk patients.
- 3 Identify key variables associated with infection risk.
- 4 Evaluate the contribution of these variables to prediction performance.
- 5 Compare the performance of different predictive models and select the most accurate and reliable approach.

Final scope: postoperative SSI prediction at the end of surgery.

The data: Assuta Ramat HaHayal EHR

35,925 surgical events

141 · 0.39% Positive SSI

244 → 97 Columns after cleaning

Figure 2- the outcome is very rare

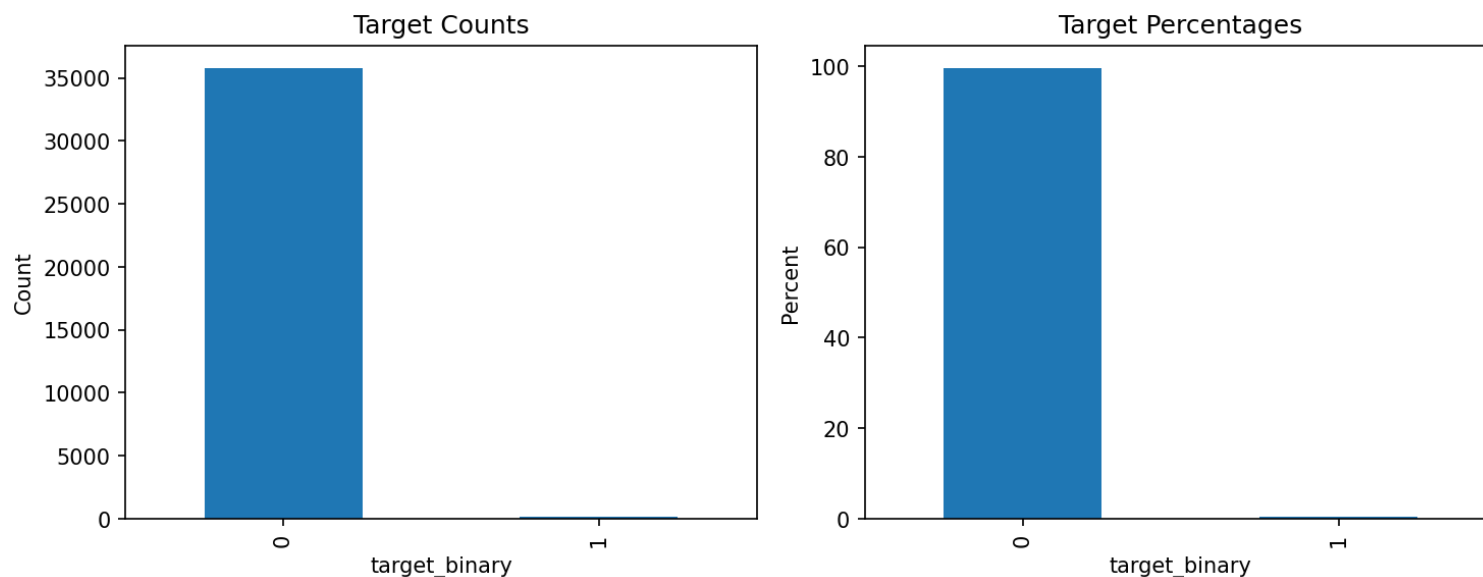
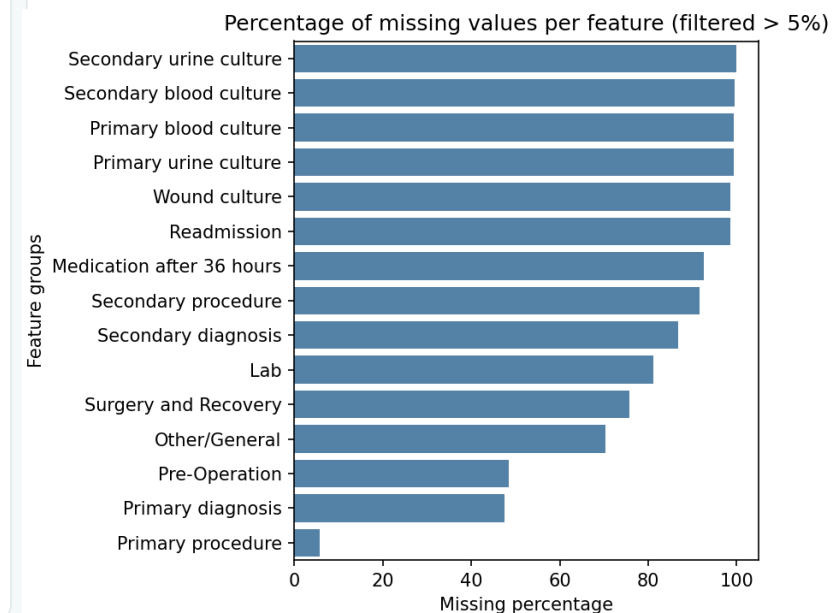
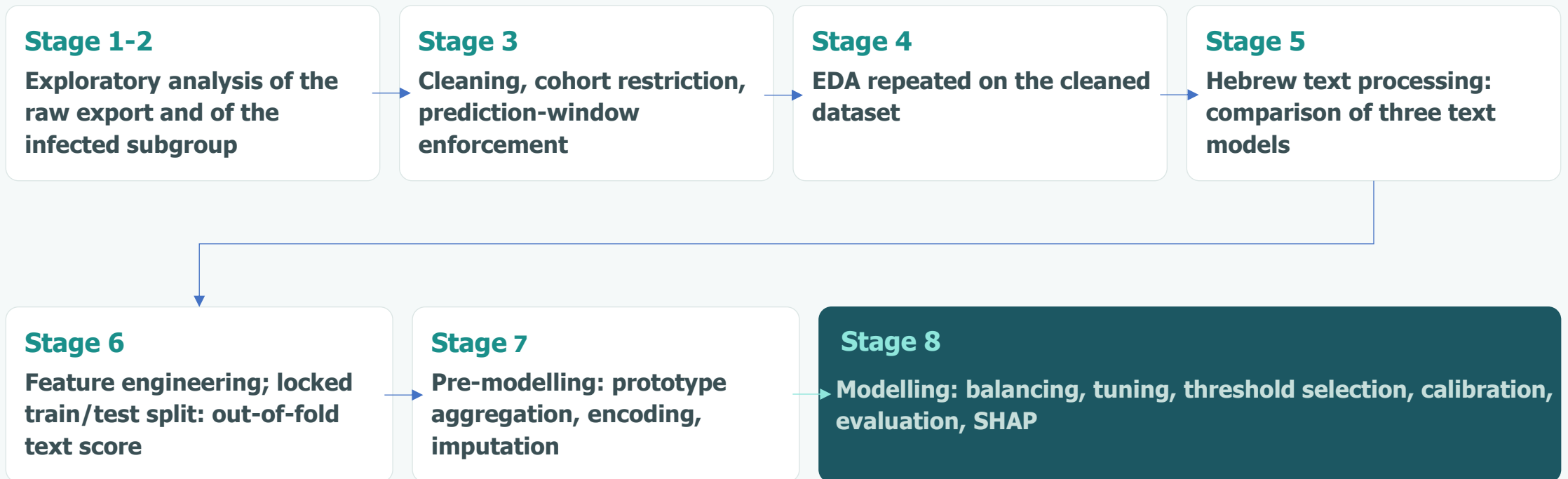


Figure 3- missingness by feature group



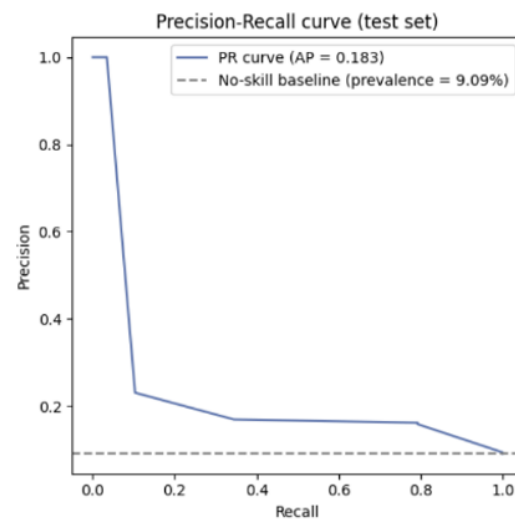
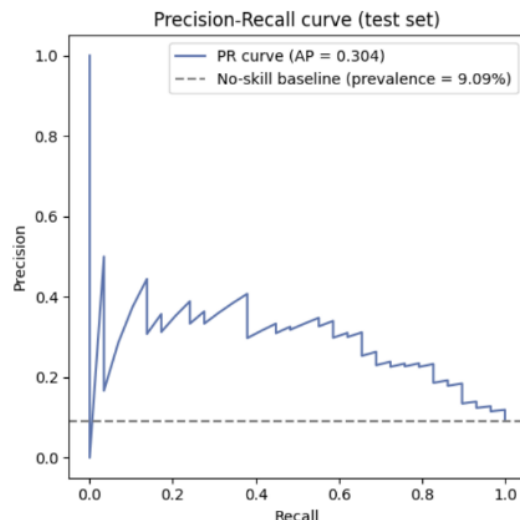
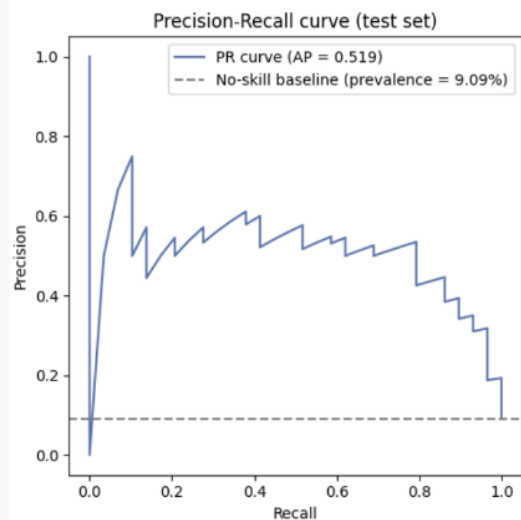
From raw export to a risk model



Comparing the text models

Method	ROC-AUC	PR-AUC (AP)	Precision	Recall
TF-IDF + logistic regression	0.933	0.519	0.577	0.517
TF-IDF + XGBoost	0.821	0.304	0.286	0.069
Llama 3.3:70b, prompted	0.701	0.183	0.169	0.345

Precision–recall by representation



Evaluated on a rebalanced 319-note split (29 positive), built to rank the three representations

The simplest model won decisively over a 70-billion-parameter LLM

SECOND COMPARISON

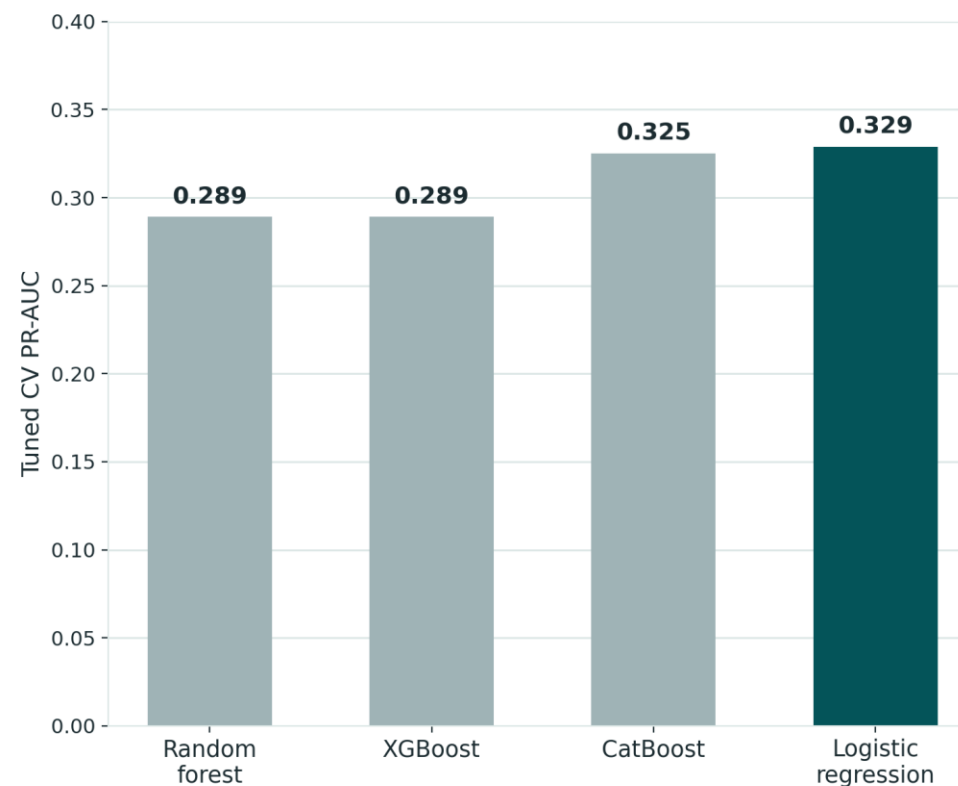
Comparing the tabular models

Model	Tuned CV PR-AUC	Tuning runtime (s)
Logistic regression - locked	0.329	11
CatBoost	0.325	1,547
XGBoost	0.289	31
Random forest	0.289	86

Takeaway

Logistic regression and CatBoost were a near-tie on CV PR-AUC. The tie was resolved for the cheaper, more interpretable SMOTE-balanced linear model.

Tuned CV PR-AUC by model



Final model performance

88.1% recall- 37/42 caught

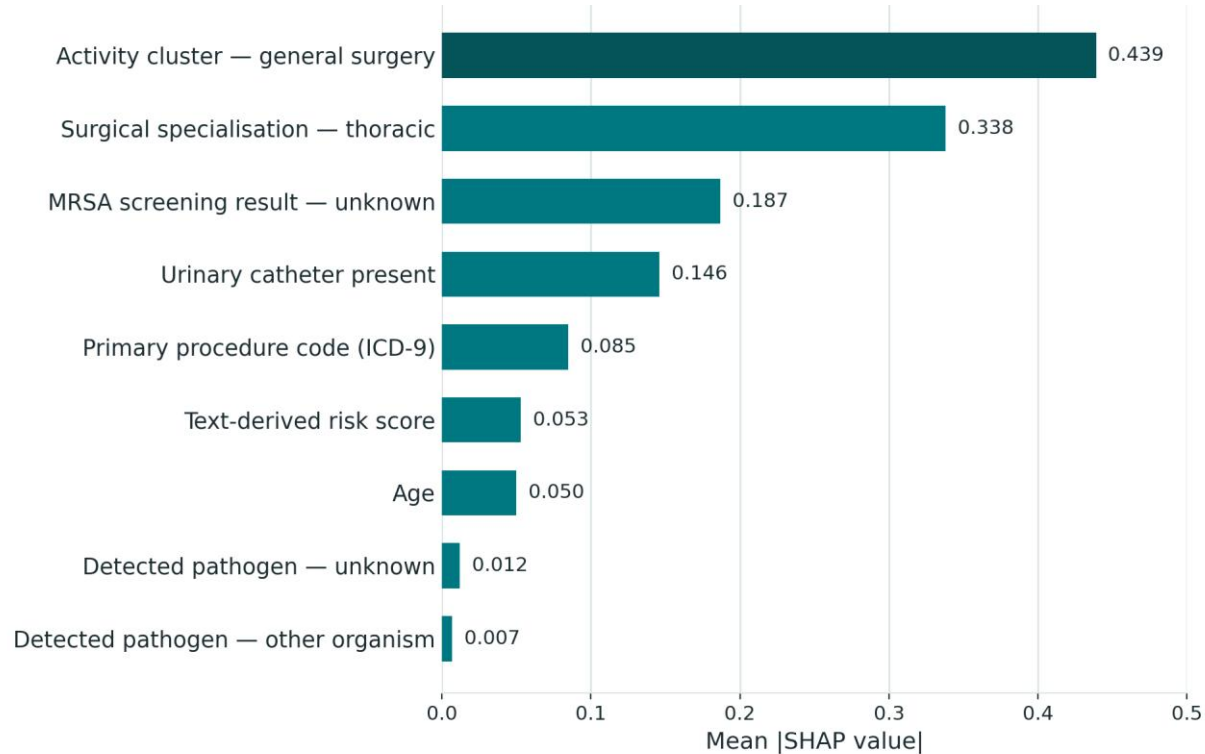
4.7% accuracy

0.183 PR-AUC

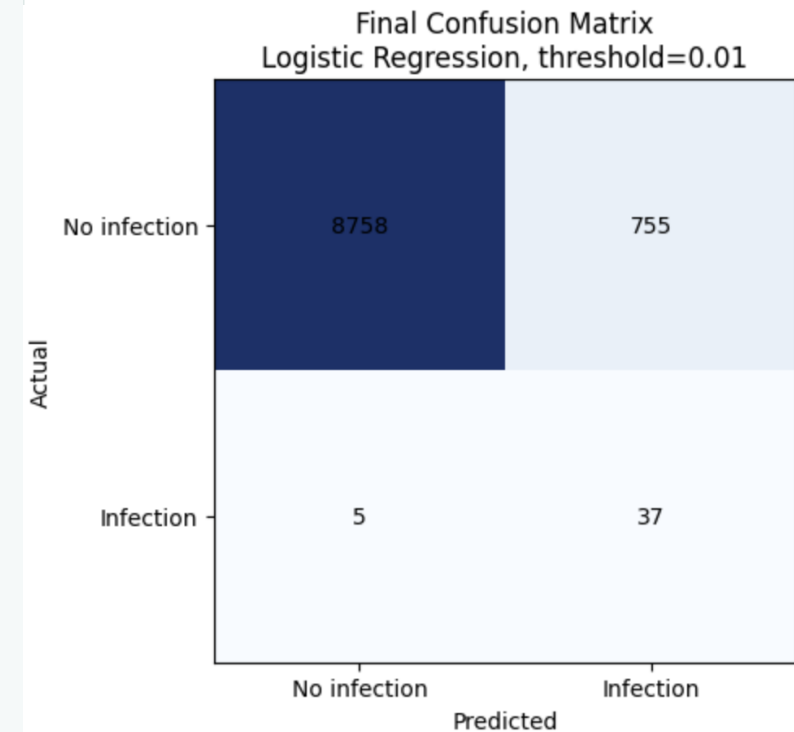
21.4 NNS

0.0039 Brier score- calibration

What drives the score (SHAP)



Confusion matrix - 9,555 events, 42 infections



Clinical workload: 82.9 alerts per 1,000 procedures.

Conclusions

- 1 Hebrew clinical narrative carries genuine infection**

- 2 High discrimination does not mean high precision**

- 3 The prediction point is a clinical decision, not a technical parameter**

- 4 A single scalar may be too narrow a channel for multimodal integration**

- 5 Leakage control has to be designed into the pipeline**

Objectives: 4 achieved / 1 partially achieved- early identification demonstrated retrospectively, not prospectively.

Limitations

Statistical power

42 positive cases in the test set.

A compressed feature set

266 features were reduced to 20 before modelling.

Pre-operative by clinical definition

No patient-level timestamp

Possible cohort artefact

Four zero-infection activity clusters were excluded

Limited post-discharge follow-up

Infections outside Assuta missed

Performance varies by subgroup

Large subgroup differences

Directions for further work

Richer text integration

Beyond one scalar score

Hebrew Transformer

Domain-adapted, GPU-trained

Subgroup-specific thresholds

Especially thoracic surgery

Prospective evaluation

On newly admitted patients

Dynamic risk model

Continuously updating

Hospital dashboard

Surface score in workflow

Thank you!

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Special thanks to:

Dr. Manor Shpriz- Clinical Supervisor, Assuta Ramat HaHayal

Galina- Infection Control Nurse, Assuta Ramat HaHayal

David Falka- System Team, Information Systems Division, Assuta

